

MAGNETOM

Replacement of Parts System

Patient Handling

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR:
from Impact/Expert
from Vision

Document Version

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The table lists the adjustments required after replacing individual boards or assemblies.



If the work is being performed while the magnet is not energized (e.g. when the system is being transported into the building), standard service equipment may be used.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. direct position drive) and the protective earth wire connected to the "main component" (e.g. patient table).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Patient table				
Assembly	Time	Adjustments required	Final checks	Required Tools
Table electronics (for C13-Spectroscopy customers a different controller board is installed)	5 min	No adjustments required	Verify all table movements according to the user manual	none
Frequency converter	15 min	Run parameter adjustment program	Verify vertical table movements	none
Display	15 min	Adjust the contrast and brightness if necessary	none	Non-magnetic tool kit
Spindle	1.5 hrs. (second person recommended)	Adjust the switches	Verify vertical table movements	Non-magnetic tool kit
Guide rails	7 h	- The magnet must be ramp down before starting the exchange of the guide rails - Adjust the switches	Verify horizontal table movements	Non-magnetic tool kit and Allen key 12 mm (included in torque wrench accessory kit), MPSU 3600
Lifting motor	45 min	- Position the new motor on the plate according to the original motor position. Only the first few systems (app. 50) have a different motor position. - Check the belt tension	Verify vertical table movements	Non-magnetic tool kit
Lifting motor (Mobiles)	8.0 h	- For mobiles the magnet must be ramp down before starting the exchange of the vertical motor - Check the belt tension	Verify vertical table movements	Non-magnetic tool kit MPS K2200
Vertical tooth belt	20 min	Check the belt tension	Verify vertical table movements	Non-magnetic tool kit

Patient table				
Assembly	Time	Adjustments required	Final checks	Required Tools
Horizontal motor assembly	30 min	No adjustments required	Verify horizontal table movements	Non-magnetic tool kit
Horizontal tooth belt	1 h	No adjustments required	Verify horizontal table movements	Non-magnetic tool kit
Horizontal belt drive	2 h	Check the belt tension	Verify horizontal table movements	Non-magnetic tool kit
Guide rollers	30 min	No adjustments required	Verify horizontal table movements	Non-magnetic tool kit
Tooth belt of the patient table	2 h	Check the belt tension	Verify horizontal table movements	Non-magnetic tool kit
Carrier panel of the table top	2 h (two persons)	Check the belt tension	Verify horizontal table movements	Non-magnetic tool kit
Table top	2 h (two persons)	Check the belt tension	Verify horizontal table movements	Non-magnetic tool kit
RF-Cable	3.0 h	No adjustments required	- Verify horizontal table movements - Quality Assurance/ SN_DIP	Non-magnetic tool kit

Note:

The RF wave trap cannot be replaced. The entire tabletop must be replaced. The RF Cable can only be replaced in pairs.

PMU				
Assembly	Time	Adjustments required	Final checks	Required Tools
PMU converter	5 min	No adjustments required	PMU Test with the service SW	Non-magnetic Screwdriver

PMU				
Assembly	Time	Adjustments required	Final checks	Required Tools
PMU Front end	15 min	No adjustments required	PMU Test with the service SW	none
PMU display	15 min	No adjustments required	PMU Test with the service SW	Non-magnetic tool kit

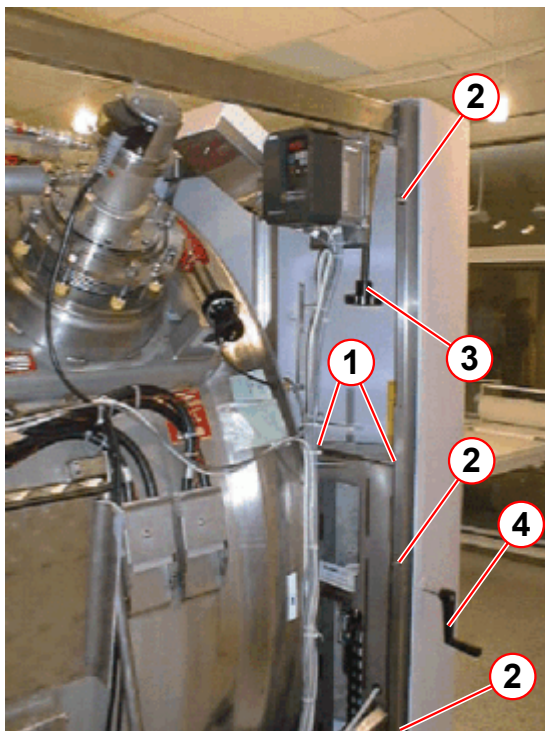
Light localizer				
Assembly	Time	Adjustments required	Final checks	Required Tools
Light localizer assembly	20 min	Calibration required	n/a	Non-magnetic tool kit



When replacing parts follow all general safety guidelines and the special safety guidelines indicated in the replacement instructions.

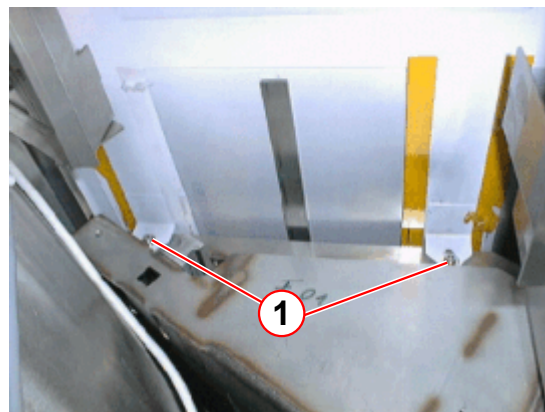
2.1 Removing the column cover

Fig. 1: Column cover remove



- (1) Screws
- (2) Allen screws
- (3) Hand wheel
- (4) Hand crank

Fig. 2: Guide cover plate_remove



- (1) Screws

2.1.1 Required Tools

- Non-magnetic tool kit

2.1.2 Preliminary Steps

- Remove the side covers

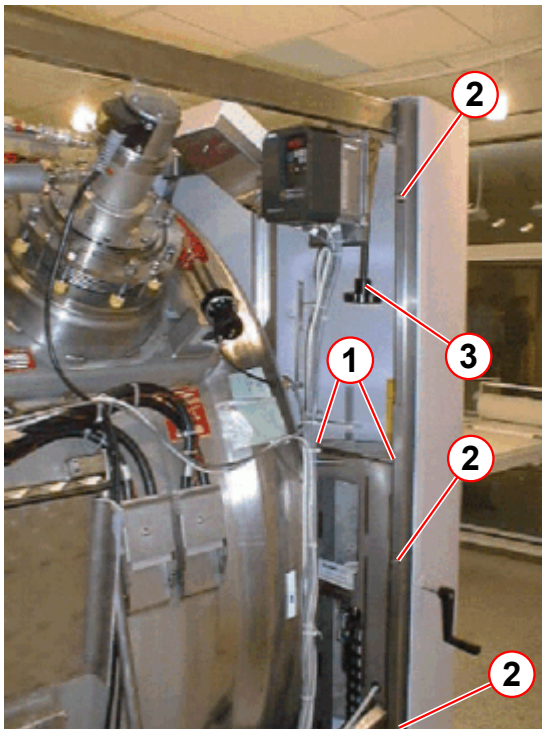
2.1.3 Replacement

- Remove the PMU-Display (if applicable). To remove the PMU-display switch off the F15/RF-Room in the CCA cabinet.
- Move the table to the uppermost position.
- Using the hand crank, crank the table further up.
- Remove both guide cover plates with the two screws (→ 1/ Fig. 2 Page 10) (access from the back).
- Using the hand crank, crank the table back into the electronic travel range.
- Move the table to the lowest position.

- Using the hand crank, crank the table further down until it reaches the mechanical end stop.
- Remove the side cover of the magnet by removing the three screws (→ 2/Fig. 1 Page 10).
- Remove the hand crank.
- Rotate the adjustment wheel to raise the cover.
- Remove the cover (lift it up off the support arm).

2.2 Mounting the column covers

Fig. 3: Column cover mount



- () Screws
- () Allen screws
- () Hand wheel

2.2.1 Required Tools

- Non-magnetic tool kit

2.2.2 Preliminary steps

- The patient table should be in the lowest position.
- Using the hand crank, crank the table down until it reaches the mechanical end stop.

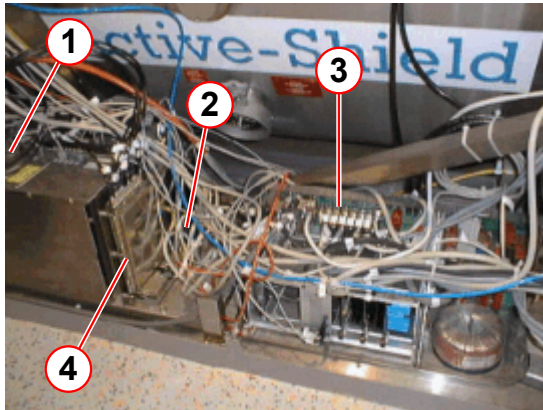
2.2.3 Mounting

- Lift the cover over the table support arm into position.
- Rotate the adjustment wheel to lower the cover. Make sure that the cover is not resting on the kick plate.
- Attach the three Allen screws (→ 2/Fig. 3 Page 12) of the front cover.
- Crank the table up into the electronic travel range.
- Move the table back up into its uppermost position.
- Using the hand crank, crank the table up.

- Attach both cover plates with the two screws (→ 1/Fig. 3 Page 12) for the table support arm slots.
- Attach the decorative blind (the spring must be under tension - 40 revolutions) with both retaining plates underneath the table support arm.
- Using the hand crank, crank the table until it reaches the electronic travel range.
- Reattach (if applicable) the PMU display. (Before reattaching, switch off F15.) Align the display with the front cover.
- Reattach the side cover of the magnet.

2.3 Removing / mounting the guide rails

Fig. 4: Table electronic rack



- (1) Tales
- (2) Connector plate
- (3) Table electronics
- (4) LCCS

Fig. 5: Grease supply



- (1) Grease supply line

2.3.1 General

The table is moved in the vertical direction via a docking plate above two guide rails.

The guide rails and the docking plate must be adjusted to each other.

During replacement of guide rail (Siemens part no. 4763004 ; Trumpf part no. 1239492) only the rails and the related roller cassetts have to be exchanged.

The docking plate including attachments will be re-used and is not available as spare part either.

Insert the guide cassetts and rails according and watch out for parallel mounting.

2.3.2 Required Tools

- Non-magnetic tool kit
- 12 mm non-magnetic Allen key (included in torque wrench accessory kit, order no.: 08396116)
- MPSU 3600

2.3.3 Preliminary Steps

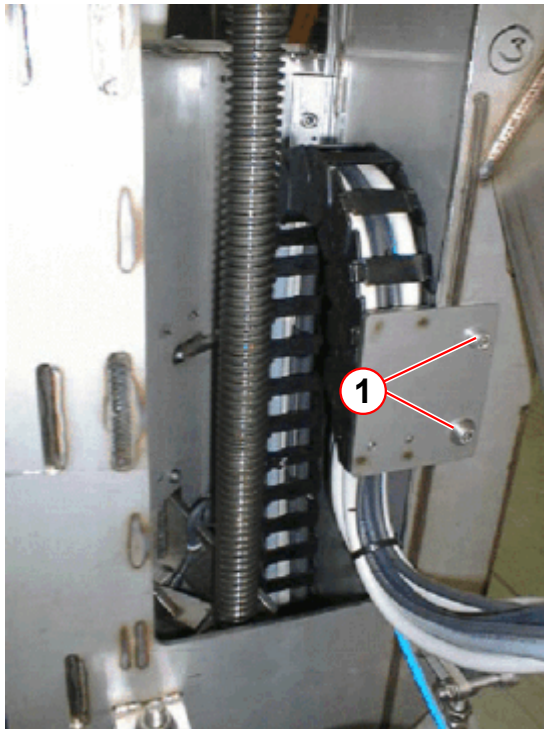
- Ramp down the magnet (Refer to Start Up Instruction).

- Switch the F15/RF-Room off in the CCA cabinet.
- (→ Removing the column cover / Page 10)

2.3.4 Removing

- Disconnect the following cables:
 - X30, X31, X32, X22 and X2 (PMU option) at the table electronics
 - air control line connection (blue hose)
 - X1 and X2 at the connector plate (→ 2/Fig. 4 Page 14) between the table electronics and LCCS
 - X11 and X12 on the LCCS
 - X6 on the TALES (at the back)
 - grease supply line (→ 1/Fig. 5 Page 14)

Fig. 6: Back of chain mounting



(1) Screws

Fig. 7: Chain labeling



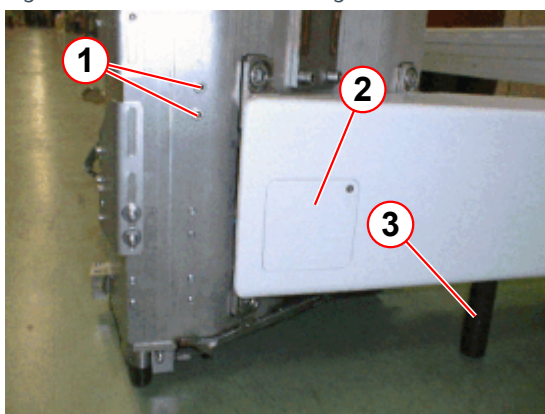
(1) Label

- Remove the attachment screws (→ 1/Fig. 6 Page 15) for the chain



Label the cable exit from the chain (e.g. with scotch tape or waterproof pen)

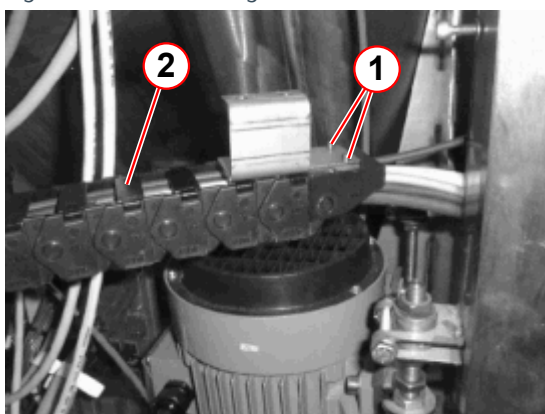
Fig. 8: Front of chain mounting



- (1) Screws
- (2) Cover plate
- (3) Support bolts

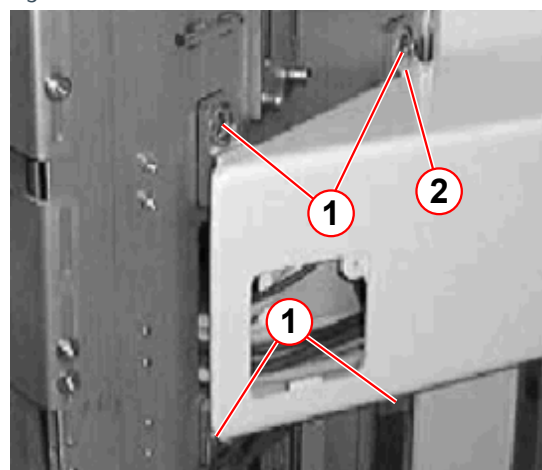
- Remove the attachment screws (→ 1/Fig. 8 Page 16) for the chain mounting.
- Remove the cover plate (→ 2/Fig. 8 Page 16) on the table arm.

Fig. 9: Chain mounting



- (1) Screws
- (2) Clips

Fig. 10: Table attachment



- (1) Allen screws
- (2) Locking pin

- Pull the chain out toward the back.
- Remove the two screws from each mounting piece (→ 1/Fig. 9 Page 16) at the chain (front and back).
- Open the clips (→ 2/Fig. 9 Page 16) and remove the cable from the chain.



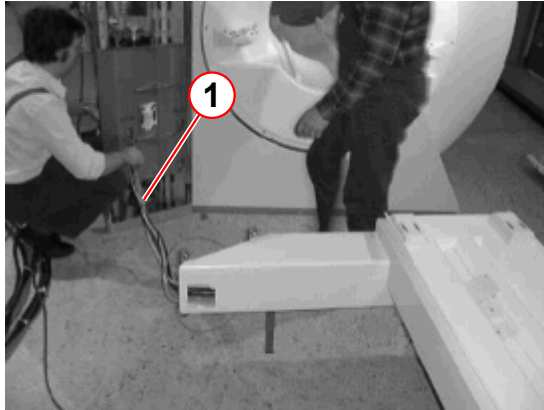
The clips can easily be damaged when opening them.

- Attach the support bolts (→ 3/Fig. 8 Page 16) (3 pcs.) to the table from underneath.
- Move the table into the lowest position using the hand crank until it contacts the spacers.
- Remove the table mounting (→ 1/Fig. 10 Page 16) (4x14 mm Allen screws, a 12 mm Allen key is required). Pull the table toward the front.



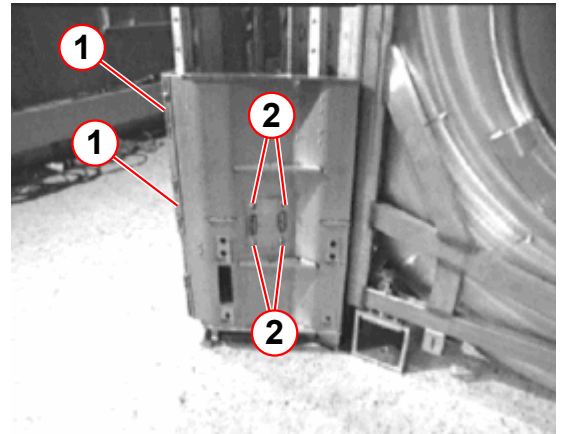
One locking pin (→ 2/Fig. 10 Page 16) is located at each of the upper mounting points of the table.

Fig. 11: Remove the table



(1) Cables

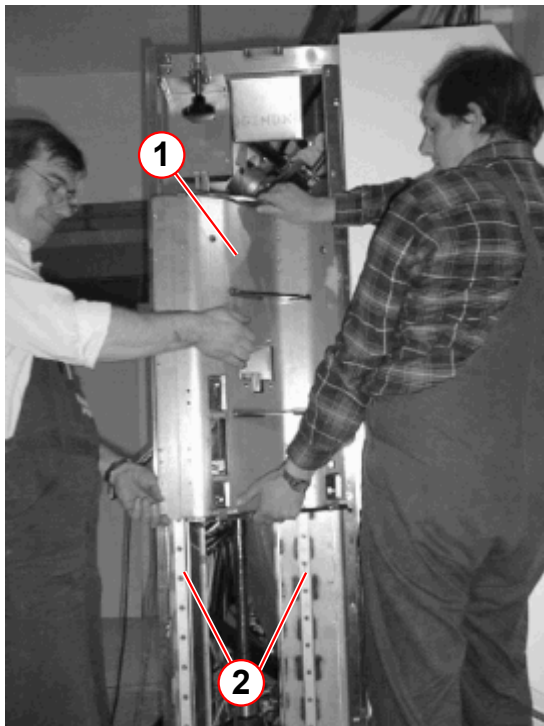
Fig. 12: Table attachment



(1) Switch plates
(2) Screws

- Pull all the cables out one at a time through the opening (→ 1/Fig. 11 Page 17).
- Remove the spindle nut mounting (→ 2/Fig. 12 Page 17) (4 screws) and withdraw it toward the front.

Fig. 13: Docking plate



(1) Docking plate
(2) Guide rails

- Remove the docking plate (→ 1/Fig. 13 Page 17) from the top.
- Exchange both guide rails (→ 2/Fig. 13 Page 17).

- Mount the spindle nut mounting (→ 2/Fig. 12 Page 17) with the 4 screws.

- Feed back all cables individually through the opening (→ 1/Fig. 11 Page 17).
- Push the table against the docking plate.
- Attach the table with 4x14mm Allen screws (→ 1/Fig. 10 Page 16) to the docking plate (using a 12 mm Allen key).
- Restore the following connections (→ Fig. 4 Page 14), (→ Fig. 5 Page 14):
 - Grease supply line
 - X30, X31, X32, X22, X95 and X2 (PMU option) at the table electronics
 - Air control line connection (blue hose)
 - X1 and X2 at the connector plate between the table electronics and the LCCS
 - X6 on the TALES (at the back)
- Insert the cables into the energy chain and close the clips.



The clips can be easily damaged when opening them.

- Attach the brackets on the energy chain (front and back) with 2 screws each.
- Pull the cables in the chain out up to the labeling. Move the chain back to its original position.
- Attach the back brackets for the chain (→ Fig. 6 Page 15).
- Attach the front brackets for the chain (→ Fig. 8 Page 16).
- Attach the cover plate to the table arm (→ 2/Fig. 8 Page 16).
- Move the table up until the support bolts (→ 3/Fig. 8 Page 16) (3 pcs.) can be removed using the hand crank.
- (→ Mounting the column covers / Page 12).
- Ramp the magnet back to field (Refer to Start Up Instruction).

2.4 Replacing the spindle

2.4.1 General

If the patient table is removed during installation it is important that the table lift mechanism is properly aligned with keyed brass piece when it is reattached.

It is possible to have the table appear to function normally while it is incorrectly riding on top on the brass spindle screw. This obviously could present a situation if the parts suddenly line up later during operation. For proper alignment, please refer to adjustment.

Although the replacement can be performed by one person, we recommend it be done by two people.

2.4.2 Required Tools

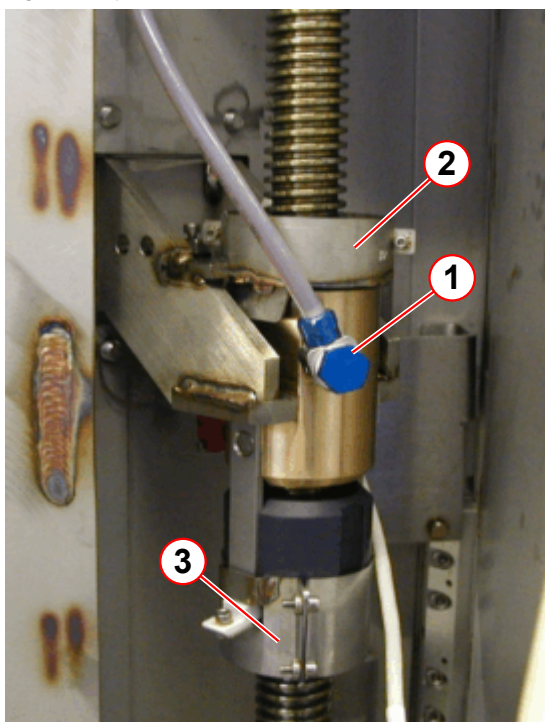
- Standard MR tool kit
- Manual crank, material number 04765231
- Service bolt, long material number 4765801, and service bolts short, material number 4765793
- Leather work gloves for holding the motor spindle
- One piece of 3' 4' lumber 60cm long (24')
- Non-magnetic gauge (e.g. 2/3mm non-magnetic metal...)

2.4.3 Preliminary Steps

- Switch off F15 (CCA)
- Remove the column cover
- Remove the splash protection panel

2.4.4 Removing the spindle

Fig. 14: Spindle nut



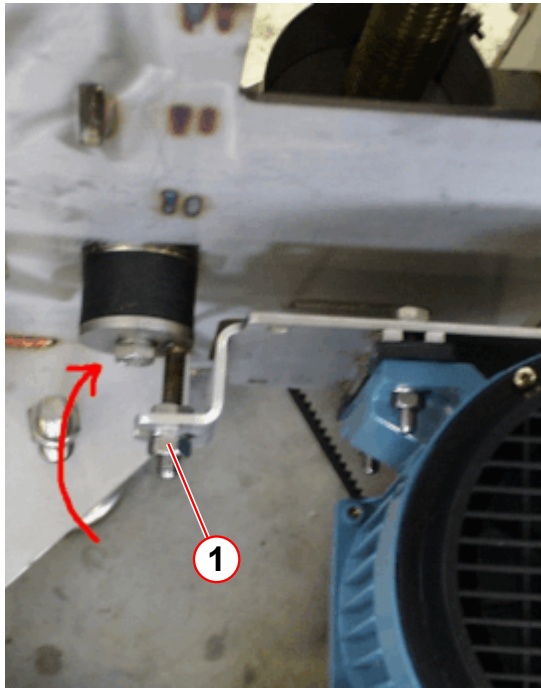
- (1) lubricant hose
- (2) upper splash protection
- (3) bottom splash protection

- Table position: 1/3 from the uppermost position to gain better access for the subsequent steps.
- Remove the lubricant hose including the connection (→ 1/Fig. 14 Page 20).
- Take off the upper splash protection (→ 2/Fig. 14 Page 20) by first removing 2 screws on the front covers. Next, take off the back cover by removing one screw.
- Disassemble the bottom splash protection (→ 3/Fig. 14 Page 20) and use it for the new spindle.



The covers (→ 2/Fig. 14 Page 20) will still be needed for the new spindle.

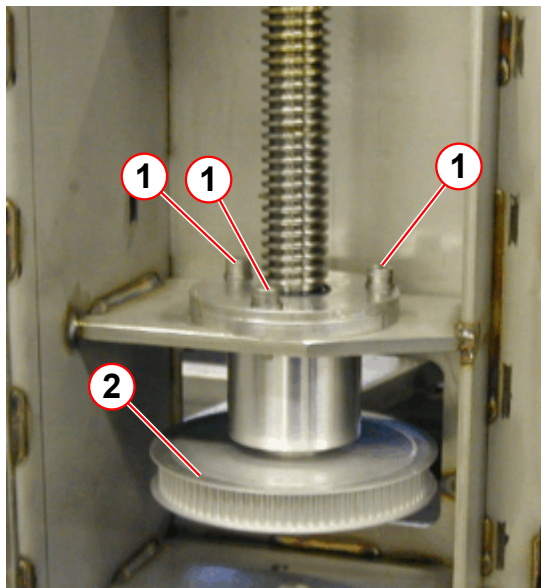
Fig. 15: Belt



(1) lock nut

- Table position: 2/3 from the uppermost position to gain better access for the subsequent steps.
- Use the lock nut (→ 1/Fig. 15 Page 21) to loosen the belt on the vertical motor mount.
- Loosen the lock nut (→ 1/Fig. 15 Page 21) in order to remove the belt.

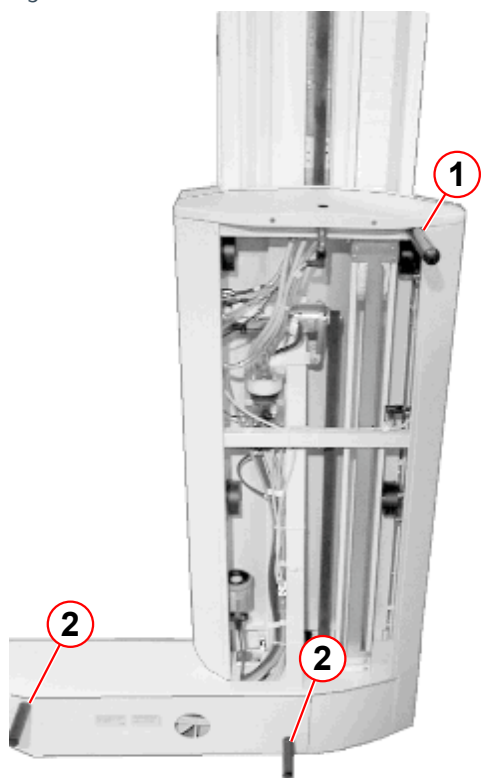
Fig. 16: Belt pulley



(1) Screws
(2) Belt pulley

- Remove the belt pulley (→ 2/Fig. 16 Page 21) on the lifting spindle using an 13mm spanner (pull it off toward the bottom).
- Remove the three screws on the lower spindle bearing (→ 1/Fig. 16 Page 21).

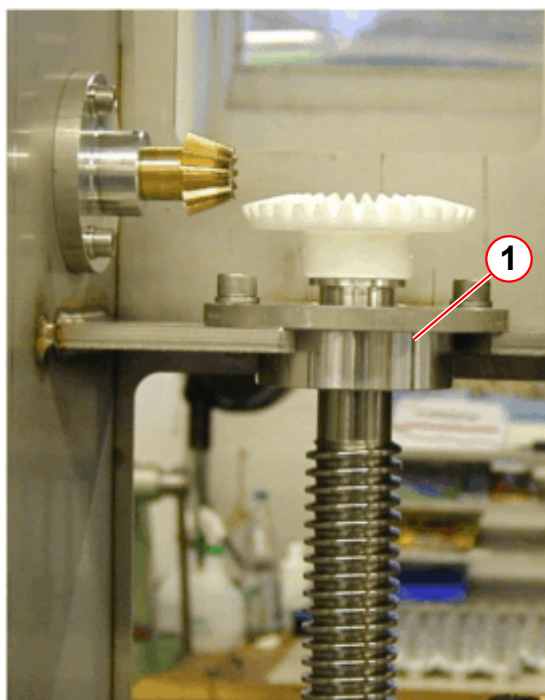
Fig. 17: Table service bolts



- (1) Service bolt long
(2) Service bolts short

- Attach the support bolts (3 pcs.) to the table from underneath.
- Use the hand crank to lower the table until there is no load on the spindle nut.

Fig. 18: Upper spindle bearing



- (1) upper spindle bearing

- Remove the two screws on the upper spindle bearing (→ 1/ Fig. 18 Page 22).

Fig. 19: Spindle nut mount



(1) Spindle nut mount

- Remove the spindle nut mount (→ 1/ Fig. 19 Page 23) by removing the 4 screws (these are secured with loctite).



Ensure that switch SY6 is not damaged when removing the spindle nut mount. The cable for the SY6 switch is secured as a loop in the suspension arm.

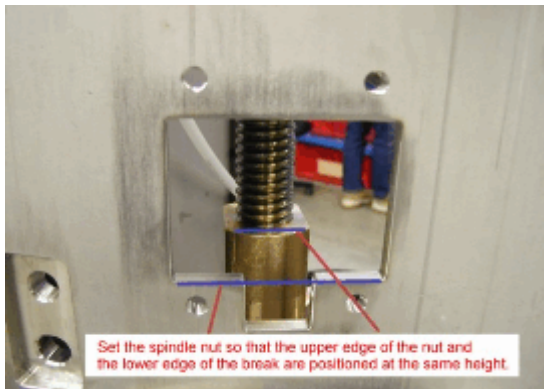
- Remove the entire spindle including bearings from the top.



The spindle bearings are slightly magnetic. Keep distance to the magnetic field.

2.4.5 Installing the spindle

Fig. 20: Installing the spindle nut mount



- Attach the lower splash protection to the new spindle.
- Insert the new spindle unit from the top.
- Press the upper and lower spindle bearing to the back (toward the magnet) and tighten the screws loosely for the time being.
- Place the spindle nut in the correct position (→ Fig. 20 Page 23).



The position of the spindle nut can be changed by rotating the nut and at the same time holding the spindle so it is locked.

Ensure that the inlet for the lubricant is facing to the back (toward the magnet).

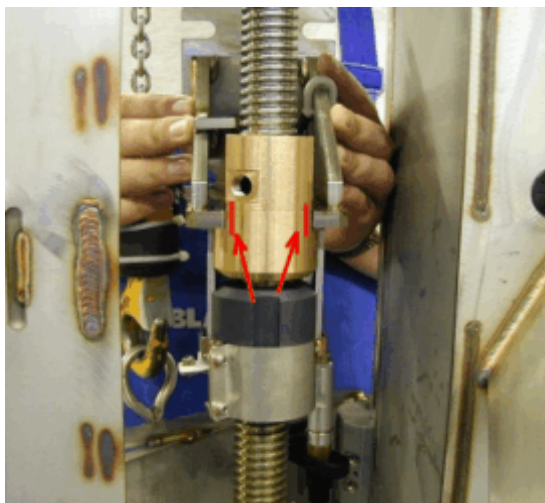
- Insert the spindle nut mount and fasten the 4 screws loosely using loctite.

2.4.6 Adjustment



If the spindle is not properly aligned, e.g. after replacement, it may lead to high friction and sporadic switch-off of the vertical drive.

Fig. 21: Installing the spindle nut mount



- Adjust the spindle nut mount so that the nut has the same gap on the left and right side.
- Use the hand crank to raise the table until the spindle nut has a full load.



The mount to the spindle nut is thereby balanced proportionately.

- Tighten the 4 screws of the spindle nut bracket.

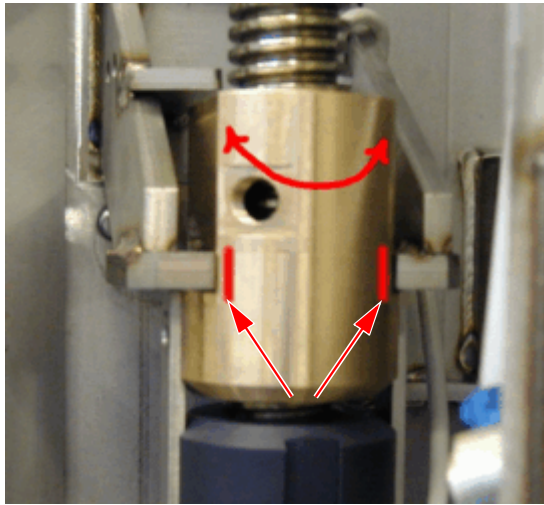


To maintain the position of the mount, tighten the screws in increments of 1/4 turns and diagonally.

- Tighten the screws on the upper and lower spindle bearing (→ 1/Fig. 16 Page 21), (→ 1/Fig. 18 Page 22).
- Place the toothed belt pulley (→ 2/Fig. 16 Page 21) on the spindle and secure it with the screw.
- Insert the toothed belt (→ 1/Fig. 15 Page 21) and turn the nut on the vertical motor to tighten the belt.

2.4.7 Verification of the gap (left and right)

Fig. 22:



- Move the table to its uppermost position.
- Remove the support bolts (3 pcs.)
- Position the wooden beam under the table arm.
- Lower the table until there is no load on the spindle nut.

2.4.7.1 Verification procedure:

- Ensure that the spindle nut can be moved within the mount (the nut moves from left to right).



The gap does not need to be the same size.

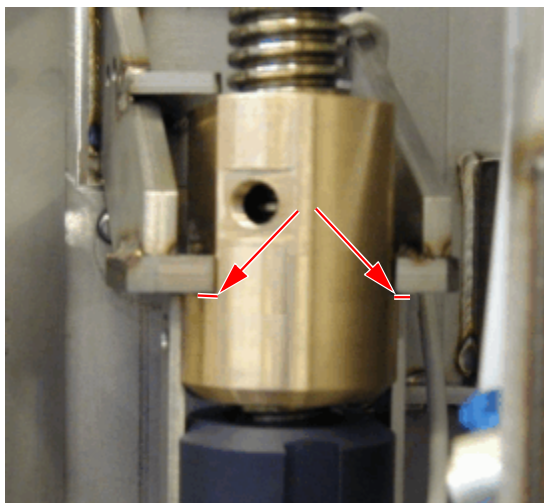
If the spindle nut cannot be moved from side to side, the upper spindle bearing (→ 1/ Fig. 18 Page 22) must be shifted in position until the spindle nut can be moved freely.



The verification procedure should be repeated until it is no longer necessary to adjust the spindle bearing.

2.4.8 Verification of the gap (below)

Fig. 23: Inserting the spindle nut mount



The table is set to the uppermost position and the spindle nut has no load (position the wooden beam under the table arm).

2.4.8.1 Verification procedure:

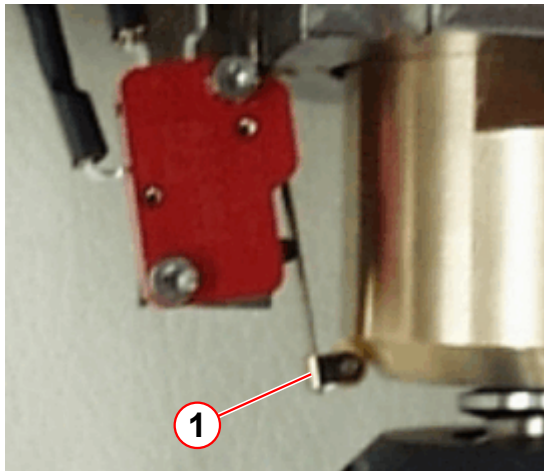
- The energy chain should be removed for this procedure in order to check the gap.
- Use a gauge to check the gap on the left and right side.
- The gap must be the same size.
- If this is not the case, loosen the spindle nut mount (→ 1/ Fig. 19 Page 23) and rotate it accordingly. When tightening the screws, refer to the Notice in (→ Adjustment / Page 24).



If the spindle nut mount was adjusted, both verification procedures must be repeated. (→ Adjustment / Page 24) followed by (→ Verification of the gap (below) / Page 26).

2.4.9 Verification of the SY6 switch position

Fig. 24: Switch SY6



(1) Switch flag

The table is set to the uppermost position and the spindle nut has no load (position the wooden beam under the table arm).



The switch should release once the switch flag (→ 1/Fig. 24 Page 27) is located in the middle of the incline. If this is not the case, reposition the switch.

2.4.10 Final steps

- Assemble the upper and lower splash protection.
- Attach the lubricant hose.
- Position the cable with the SY6 switch in the suspension arm as a loop.
- Tighten the energy chain mount.
- Attach the splash protection.
- Attach the column covers.
- Switch on F15 (CCA).
- Select **System> Control** in the main menu, the **System Manager** dialog box opens, go to the MR Scanner card, click reboot.

2.5 Replace vertical motor assembly

2.5.1 General



WARNING

Please follow the General Safety Information (→ General Safety Notes / TD00-000.860.01) as well as the MR-Specific Safety Information (→ MR-specific safety information / MR-000.860.01)!

Error to comply with the Safety Information may result in death or serious injury.

- » Read the safety notes contained in the General Safety Information as well as the MR-Specific Safety Information in detail prior to starting work. Adhere to the information provided at all times.



The vertical motor consists of magnetic parts, keep distance to the magnetic field, or ramp down the magnet. For mobiles the magnet must be ramp down.



DANGER

Vertical motor is strongly ferromagnetic!

Magnetic objects may cause injury.

- » Remove the motor only when it is swivelled out completely and hold the ferromagnetic motor with both hands while moving it away from the magnet.
- » Remove the motor directly out of the field by keeping maximum distance from the magnet!



DANGER

Ferromagnetic parts!

Magnetic objects may cause injury.

- » Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!

2.5.2 Tools and time required

Time:

90 min (two persons)

Tools:

- Loctite 221 (material number 2048874)

- Non-magnetic steel tool set (material number 7757706)

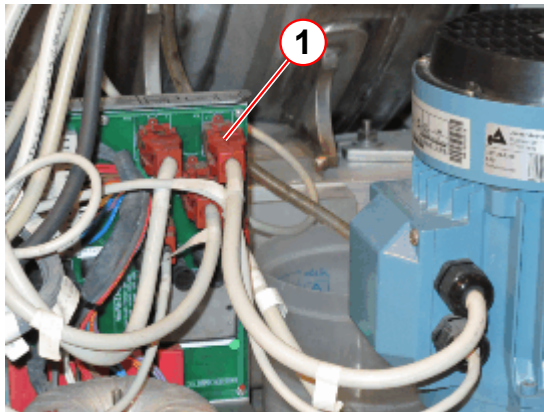
2.5.3 Preliminary Steps

- Switch OFF the system.
- Remove the left side magnet covers to reach the vertical motor.

2.5.4 Replacement

- Remove the vertical motor cable plug X36 and X3 (→ 1/ Fig. 25 Page 29).

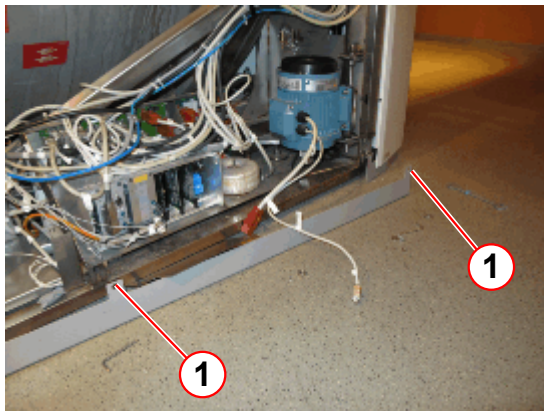
Fig. 25: Motor cable plug



(1) Plug X36 and X3

- Remove two Allen screws of the kick plate and carefully move it slightly away to have enough space to swivel out the vertical motor (→ 1/ Fig. 26 Page 29).

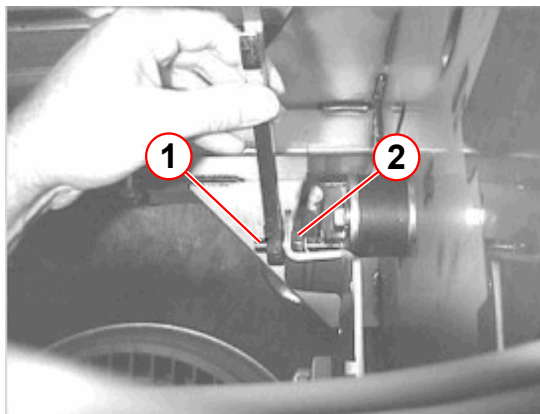
Fig. 26: Kick plate



(1) Allen screw

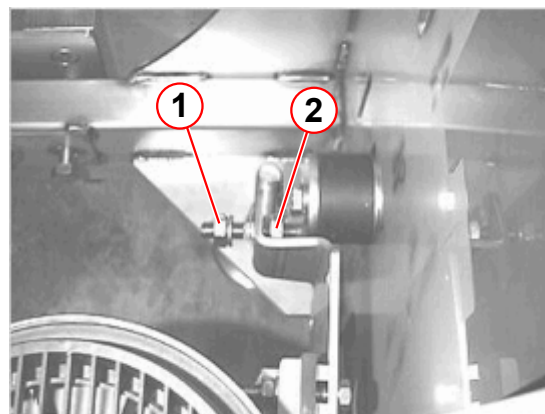
- To have a reference point for the toothed belt tension, mark with a permanent marker the position of the nut "a" and "b" (→ 1, 2/ Fig. 27 Page 30). Additionally check the toothed belt tension by hand.

Fig. 27: Belt 1



- (1) Nut "a"
(2) Nut "b"

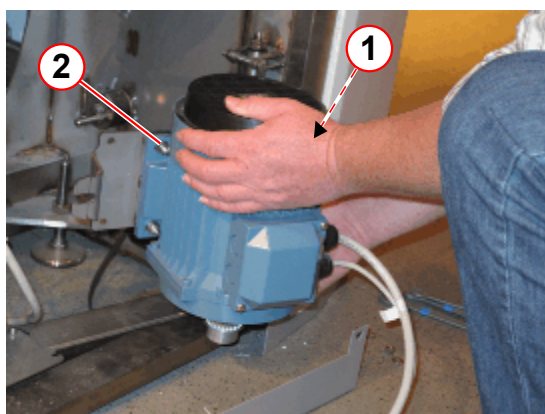
Fig. 28: Belt 2



- (1) Nut "a"
(2) Nut "b"

- Remove the nut "a" (→ 1/ Fig. 28 Page 30) for the toothed belt tension and bring nut "b" (→ 2/ Fig. 28 Page 30) to the maximum inner position. Remove the belt from the bottom side of the vertical motor.
- Swivel out the ferromagnetic vertical motor completely to have the maximum distance from the magnet.
- First remove the two nuts on the right side of the motor mounting plate (→ 1/ Fig. 29 Page 30). One person needs to hold the mounting plate in position and removes the two mounting plate nuts of the left side (→ 2/ Fig. 29 Page 30), while the other person is holding the motor with both hands. Hold the ferromagnetic motor with both hands while moving it away from the magnet, remove the motor directly out of the field and keep maximum distance from the magnet!

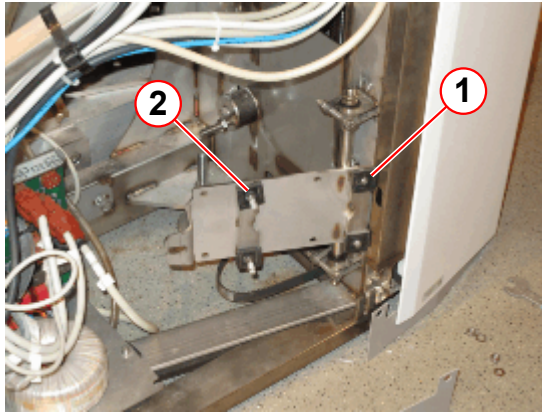
Fig. 29: Motor removal



- (1) Nuts right side
(2) Nuts left side

- When placing the new vertical motor to the mounting plate, be careful, keep maximum distance from the magnet, one person must hold the mounting plate in "swivel out" position and fixes the four mounting nuts of the new motor with Loctite 221 (material number 2048874) while the other person is holding the motor with both hands in the "swivel out" position.

Fig. 30: Mounting plate



(1) Nut right side

(2) Nut left side

- After the fixation of the vertical motor with the four nuts, swivel back the motor into home position and attach the toothed belt.
- Fix the two nuts for the toothed belt tension to the original position mark and check the toothed belt tension by hand.
- Mount the Allen screws of the kick plate.

2.5.5 Final steps

- Switch ON the system.
- Drive the patient table in vertical direction.
- Close all covers.

2.6 Replace horizontal motor assembly

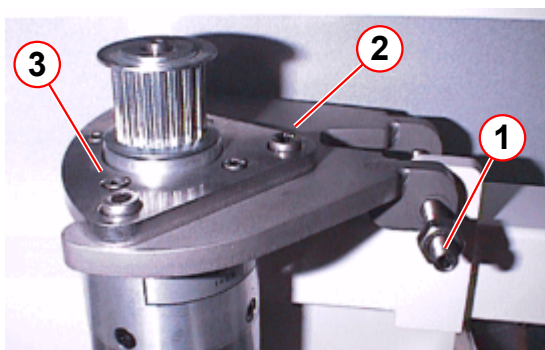


Switch off circuit breaker F15 in the CCA.

Remove the table covers. For a detailed description, refer to (→ Removable patient table / MR00-000.814.16).

- Remove the foot-end table cover (2 x M6 allen screws).
- Pull out the lower panel.

Fig. 31: Swivel plate



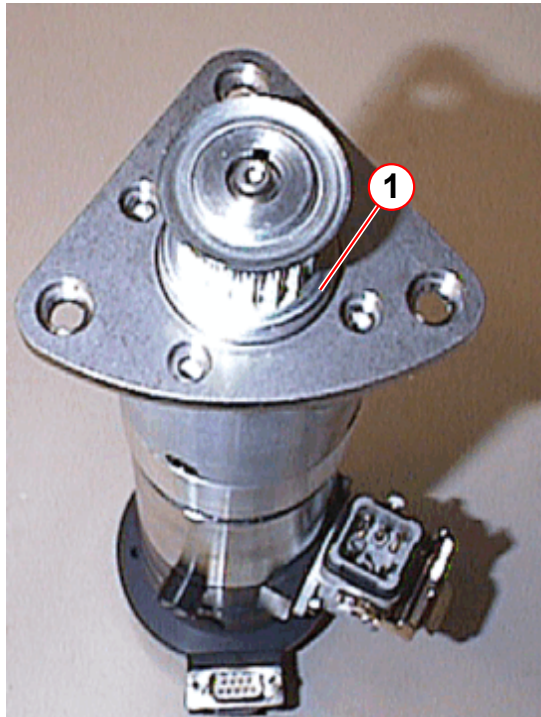
- (1) Lock nut
- (2) Swivel plate
- (3) Mounting plate

- Release lock nut (→ 1/Fig. 31 Page 32) and the M8 screw.
- Move the swivel plate (→ 2/Fig. 31 Page 32) upwards to the stop.
- Push the belt over the drive pulley.
- Remove the two connectors (one connector is locked with screws).
- Disassemble the mounting plate (→ 3/Fig. 31 Page 32) from the swivel plate (→ 2/Fig. 31 Page 32) (3 x M6 allen screws).
- Remove the motor with the incremental encoder.
- Remove the mounting plate (→ 3/Fig. 31 Page 32) from the motor (4 x M4 allen screws).



This work may be performed outside the magnet room with conventional tools, since the screws are secured.

Fig. 32: Motor assembly



(1) Mounting plate

- Attach the mounting plate (→ 1/Fig. 32 Page 33) to the new motor unit.
- Install the horizontal motor assembly in the reversed order.
- Reattach the covers in the reverse order.

2.7 Replace horizontal tooth belt

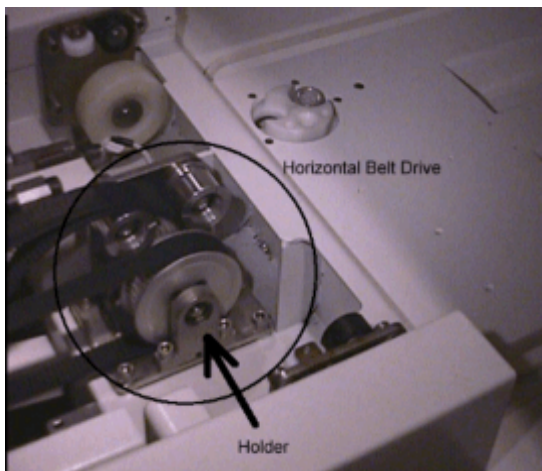


Switch off circuit breaker F15 in the CCA

Remove the table covers. For a detailed description, refer to (→ Removable patient table / MR00-000.814.16).

- Remove the foot-end table cover (2 x M6 allen screws).
- Pull out the lower panel.
- Remove the foot-end table PMU cover (4 x M5 allen screws underneath the table and 1xM5 allen screw above the table towards the magnet).
- If a PMU front end is attached
 - Remove the PMU plug from the front end.
 - Deinstall the PMU front end (3 x M4 allen screws).
- Remove the hose connections on the tabletop. (→ Fig. 34 Page 40)
- Remove the end plate (4 x M5 allen screws).
- Remove the brace plate and the stop plate (4 x M6 allen screws). (→ 1/Fig. 35 Page 40)
- Remove the four RF plug connections (baluns).
- Remove the E-chain (2 x M4 countersunk screws under the PMU Frontend) and let it hang down on the floor.
- Push the tabletop manually into the magnet until the horizontal belt drive is accessible.
- Release lock nut (→ 1/Fig. 31 Page 32) and the M8 screw.
- Move the swivel plate (→ 2/Fig. 31 Page 32) upwards to the stop.
- Push the belt over the drive pulley.
- Remove the right plate holder (2 x M8 allen screws) of the horizontal belt drive.

Fig. 33: Horizontal tooth belt



- Mount the new horizontal tooth belt.
- Reassemble in the reverse order.



You can rotate the toothed belt by 90° with your thumb and index finger. If necessary, adjust the belt tension with the screw M8. (→ Fig. 31 Page 32)

2.8 Replace the horizontal belt drive



Switch off circuit breaker F15 in the CCA.

Remove the table covers. For a detailed description, refer to (→ Removable patient table / MR00-000.814.16).

- Remove the foot-end table cover (2 x M6 allen screws).
- Pull out the lower panel.
- Remove the PMU cover (4 x M5 allen screws underneath the table and 1 x M5 allen screw above the table towards the magnet).
- If a PMU front end is attached
 - Remove the PMU plug from the front end.
 - Deinstall the PMU front end (3 x M4 allen screws).
- Remove the hose connections on the tabletop. (→ Fig. 34 Page 40)
- Remove the end plate (4 x M5 allen screws).
- Remove the brace plate and the stop plate (4 x M6 allen screws). (→ Fig. 35 Page 40)
- Remove the four RF plug connections (baluns).
- Note the number of teeth overhanging the toothed belt (the new one should be reinstalled the same way).
- Remove the foot-end interface plate for the belt (4 x M6 allen screws).
- Remove the E-chain (2 x M4 countersunk screws under the PMU frontend) and let it hang down on the floor.
- Push the tabletop manually into the magnet until the horizontal belt drive is accessible.
- Release lock nut (→ 1/Fig. 31 Page 32) and the M8 screw.
- Replace the horizontal belt drive (→ Fig. 33 Page 34)
- Pull the table with the belt at the foot-end far enough so that the foot-end interface plate can be attached.
- Attach the E-chain with 2 countersunk screws (M4 x 16).
- Route the hoses through and reconnect them.
- Reconnect the four RF plug connections.
- Route the PMU cable through if a PMU is installed.
- Attach the belt with the interface plate. The belt must overhang by the same number of teeth as prior to removal of the table. If not, the belt tension has to be adjusted.
- Install the brace plate and the stop plate (4 pcs. M6 x 16).
- After (→ Checking the belt tension of the patient table / Page 44) the covers should be reassembled in reversed order.
- Install the end plate.
- Install the PMU frontend (if applicable).
- Install the PMU cover.

- Install the lower panel and the foot-end table cover (2 x M6 allen screws) should be reassembled.

2.9 Replace tooth belt of the patient table



Switch off circuit breaker F15 in the CCA.

Remove the table covers. For a detailed description, refer to (→ Removable patient table / MR00-000.814.16).

- Remove the foot-end table PMU cover (4 x M5 allen screws underneath the table and 1 x M5 allen screw above the table towards the magnet).
- If a PMU front end is attached
 - Remove the PMU plug from the front end.
 - Deinstall the PMU front end (3 x M4 allen screws).
- Remove the hose connections on the tabletop. (→ Fig. 34 Page 40)
- Remove the end plate (4 x M5 allen screws).
- Remove the brace plate and the stop plate (4 x M6 allen screws). (→ Fig. 35 Page 40)
- Remove the four RF plug connections (baluns).
- Note the number of teeth overhanging the toothed belt (the new one should be reinstalled the same way).
- Remove the foot-end interface plate for the belt (4 x M6 allen screws).
- Remove the E-chain (2xM4 countersunk screws under the PMU frontend) and let it hang down on the floor.
- Push the tabletop manually into the magnet until the head end of the tabletop is accessible.
- Note the position of the toothed belt (the new one should be reinstalled the same way).
- Remove the head-end interface plate for the belt (4 x M6 plastic screws).
- Remove the tooth belt.
- Compare the replaced belt with the new one and make sure that the new one is the same length (if not, cut the new belt to the same length)
- Reinstall the new tooth belt.
- Reinstall the interface plate for the tooth belt. The belt should have the same position as before.
- Pull the table with the belt at the foot-end far enough so that the foot-end interface plate can be attached.
- Attach the E-chain with 2 countersunk screws (M4 x 16).
- Route the hoses through and reconnect them.
- Reconnect the four RF plug connections.
- Route the PMU cable through if a PMU is installed.
- Attach the belt with the interface plate. The belt must overhang by the same number of teeth as prior to removal of the table. If not, the belt tension has to be adjusted.
- Install the brace plate and the stop plate (4 pcs. M6 x 16).

- After (→ Checking the belt tension of the patient table / Page 44) the covers should be reassembled in reversed order.
- Install the end plate.
- Install the PMU frontend (if applicable).
- Install the PMU cover.
- Install the lower panel and the foot-end table cover (2 x M6 allen screws) should be reassembled.

2.10 Replace the table top

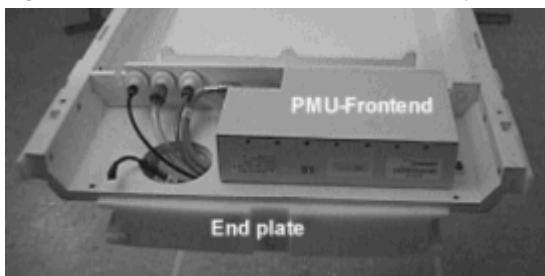


Switch off circuit breaker F15 in the CCA.

Remove the table covers. For a detailed description, refer to (→ Removable patient table / MR00-000.814.16).

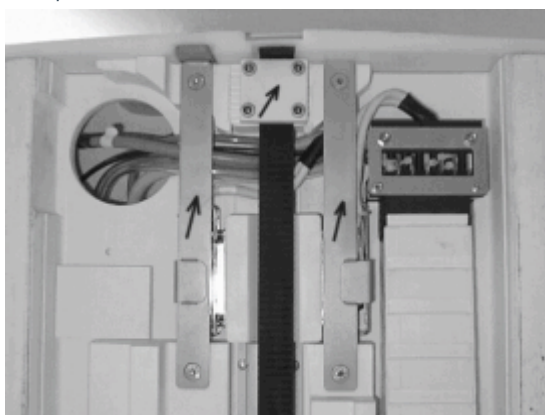
- Remove the foot-end table cover (2 x M6 allen screws).
- Remove the foot-end table PMU cover (4 x M5 allen screws underneath the table and 1 x M5 allen screw above the table towards the magnet).
- If a PMU front end is attached
 - Remove the PMU plug from the front end.
 - Deinstall the PMU front end (3 x M4 allen screws).

Fig. 34: PMU front end at standard tabletop



- Remove the hose connections on the tabletop.
- Remove the end plate (4 x M5 allen screws).

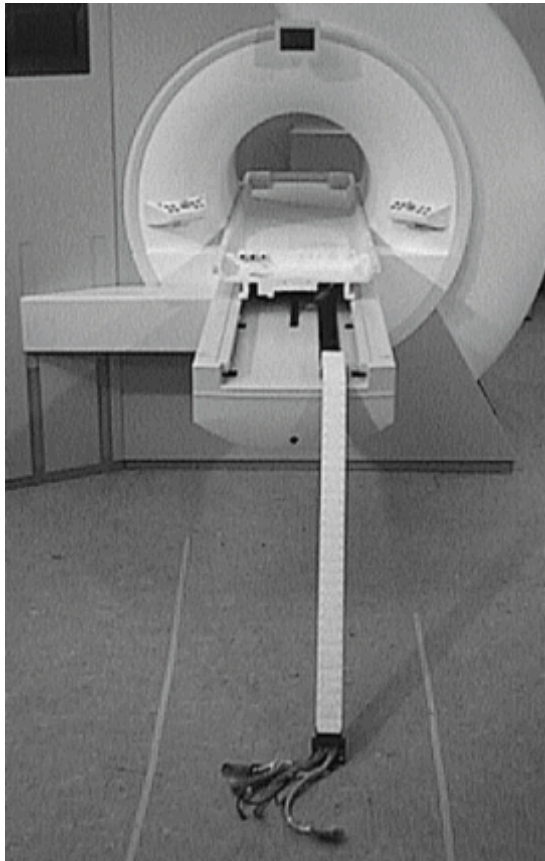
Fig. 35: Stop plate (left) and brace plate (right), Interface plate (above)



- Remove the brace plate and the stop plate (4 pcs. M6).

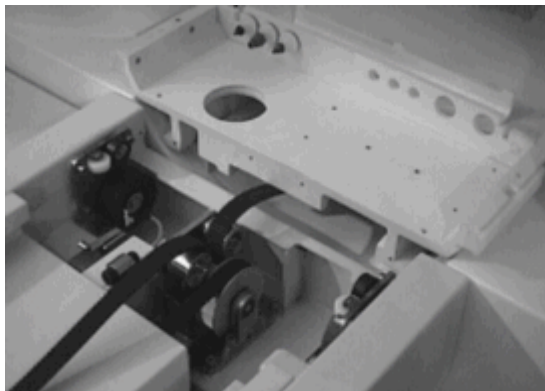
- Remove the four RF plug connections (baluns).
- Note the number of teeth overhanging the toothed belt (it should be reinstalled the same way on the new tabletop).
- Remove the foot-end interface plate for the belt (4 x M6 allen screws).

Fig. 36: E-chain removed



- Remove the E-chain (2 x M4 countersunk screws under the PMUFrontend) and let it hang down on the floor.

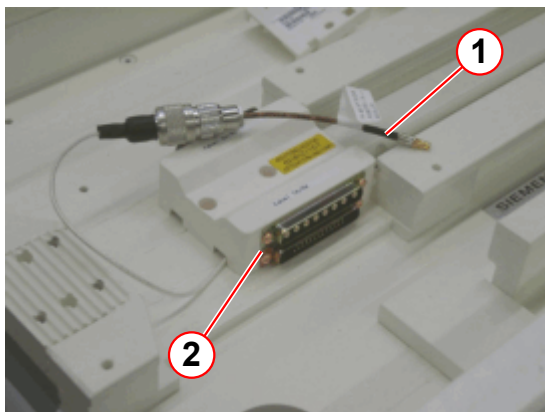
Fig. 37: Tabletop in the magnet



- Push the tabletop manually into the magnet until the head end of the tabletop is accessible.
- Note the position of the toothed belt (the new one should be reinstalled the same way).
- Remove the head-end interface plate for the belt (4 x M6 plastic screws).
- Push the tabletop further into the magnet until it disengages from the table carrier frame.
- Pull the tabletop out (2 persons required).
- Thoroughly clean the bearing surfaces in the tube, the large rollers in the table base and the bearing surfaces of the tabletop carrier with a suitable cleaner (plastics cleaner).
- Push the new tabletop into the tube (two persons required).
- Attach the belt with the interface plate. The belt should have the same position as before.

- Pull the table top with the belt at the foot-end far enough so that the foot-end interface plate can be attached.
- Attach the E-chain with 2 countersunk screws (M4 x 16).
- Route the hoses through and reconnect them.
- Reconnect the four RF plug connections.

Fig. 38: Additional coax cable has to be connect to position of the plug

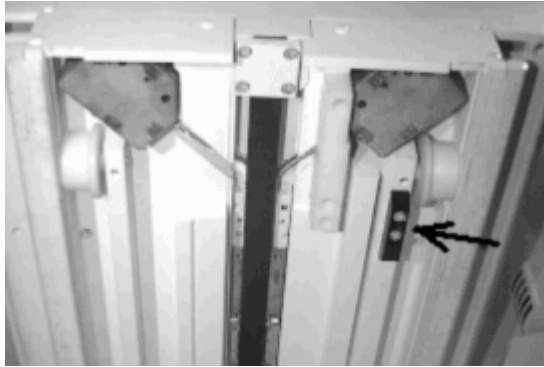


(1) Coax cable
(2) Plug

- If you have an older type of patient table the additional coax cable (→ 1/ Fig. 38 Page 42) (delivered with the table top spare part) has to be connected to the plug (→ 2/ Fig. 38 Page 42), by unscrewing the plug cover, on new type of table tops the adapter will not be used.
- Route the PMU cable through if a PMU is installed.
- Attach the belt with the interface plate. The belt must overhang by the same number of teeth as prior to removal of the tabletop. If not, the belt tension has to be adjusted.
- Install the brace plate and the stop plate (4 pcs. M6 x 16).
- (→ Adjusting the tabletop movement / Page 43)
- After (→ Checking the belt tension of the patient table / Page 44) the covers should be reassembled in reversed order.
- Install the end plate.
- Install the PMU frontend (if applicable).
- Install the PMU cover.
- Install the lower panel and the foot-end table cover (2 x M6 allen screws) should be reassembled.

2.11 Adjusting the tabletop movement

Fig. 39: Switch cam underneath the tabletop carrier



- Move the table to its uppermost position.
 - Move the tabletop by approximately 30 cm in the direction of the magnet and then back until it automatically stops.
 - The head-end of the tabletop must be flush with the table carrier; there must not be any overhang.
 - If necessary, adjust the end position switch cam as follows:
- Move the tabletop far enough into the magnet so that it reaches the switch cam underneath the tabletop carrier.
 - Adjust the cam by the same distance and in the same direction as the deviation.
 - Switch the power to the table off and back on.
 - The new table position will be sensed when the table initializes.
 - Move the tabletop again by approximately 30 cm in the foot-end direction until it automatically stops.
 - Check whether the head-end of the tabletop coincides with the tabletop carrier.
 - Repeat this procedure, until the adjustment is satisfactory.
 - The **space** between the tabletop and the magnet housing must be **within a range of 4 ± 1 mm**.

2.12 Checking the belt tension of the patient table



Switch on circuit breaker F15 in the CCA and do a "Reload/All

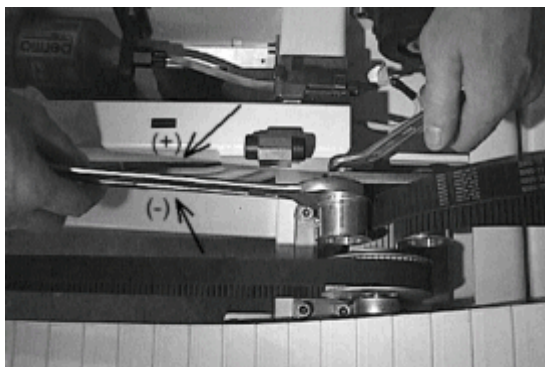
Check the correct belt tension as follows:

- Switch the laser light localizer on and attach a marker below the laser target on the tabletop.
- Switch the laser light localizer off again
- Move the table 800 mm into the magnet.
- Switch the laser light localizer back on and check the travel range at the marker. The table travel must exactly correspond to the displayed distance.
- If the travel distance is too small, the belt must be tightened again (+), otherwise it must be released (-). The accuracy should be within ± 0.5 mm. (→ Fig. 40 Page 44)
- For adjustments remove the cover plate (2 x M4 x 12 underneath and 2 x M4 x 6 Philips head).
- Adjust the belt tension by turning the eccentric pulley underneath the tabletop. (→ Fig. 40 Page 44)



A nonmagnetic 27 mm size open wrench will be delivered with the spare part.

Fig. 40: Horizontal belt drive



- Reinstall the cover plate.

2.13 Replace the carrier panel of the table top (only for removable patient table)

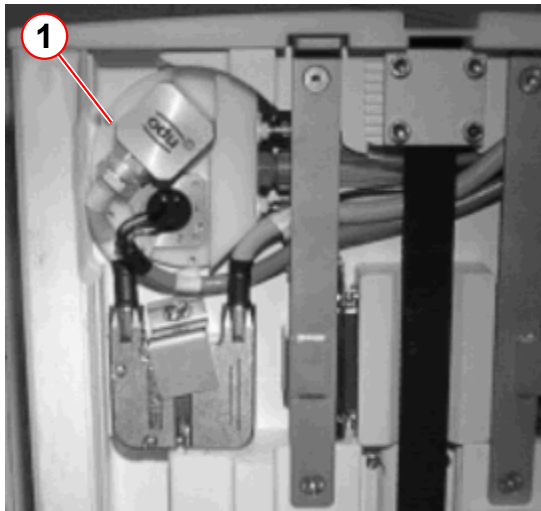
Use the trolley to remove the table top.

Perform the next steps according to the chapter (→ Replace the table top / Page 40) including the step.



The "autom. connection" has been removed from the old carrier plate and reinstalled in the new carrier plate. (→ 1/ Fig. 41 Page 45)

Fig. 41: Cable connector secured with retaining bracket under the tabletop



(1) Carrier plate

2.14 Replace the RF cable



Switch off circuit breaker F15 in the CCA.

- Remove the side covers of the magnet.

Remove the table covers. For a detailed description, refer to (→ Removable patient table / MR00-000.814.16).

- Remove the foot-end table cover (2 x M6 allen screws).
- Pull out the lower panel.
- Remove the brace plate and the stop plate (4 pcs. M6 x 16). (→ Fig. 35 Page 40).
- Remove the E-chain (2 x M4 countersunk screws under the PMU frontend) and let it hang down on the floor.
- Remove the cover plate on the table arm.
- Open the clips from the chain between patient table electronic and lifting columns and the clips of the E-chain.
- Disconnect the both RF-cables at the patient table electronic and at the front of the tabletop.
- Remove both RF-cables.



Both RF-cables must be replaced.

- Reinstall the new RF-cables.
- Reassemble in the reverse order.

3.1 Light localizer

3.1.1 Replacement

Replacing the light localizer assembly is rather simple by just removing two allen screws and disconnecting the assembly from the supply cable. Putting it back is done in reverse order. The procedure may also be used on TIM systems (→ Fig. 42 Page 47).

Fig. 42: Various light localizer designs



3.1.2 Calibration

It is the objective to align the two laser beams in perpendicular to the patient table in z, x and y direction.

The adjustment works for all systems alike as the laser assembly used is the same. Only the light localizer housing is different (→ Fig. 42 Page 47) and for the Allegra the holder and phantom are different.

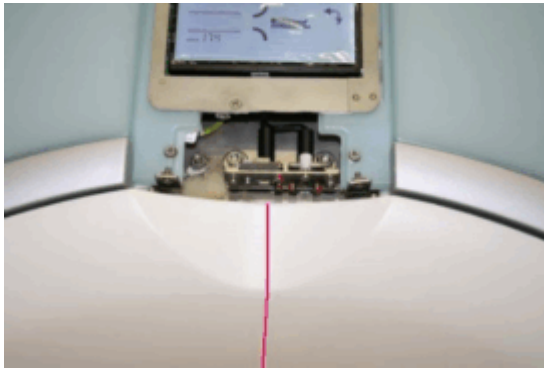
The sequence of the calibration is irrelevant - do as you like.

The procedure described hereafter is just an example.

3.1.2.1 Tips

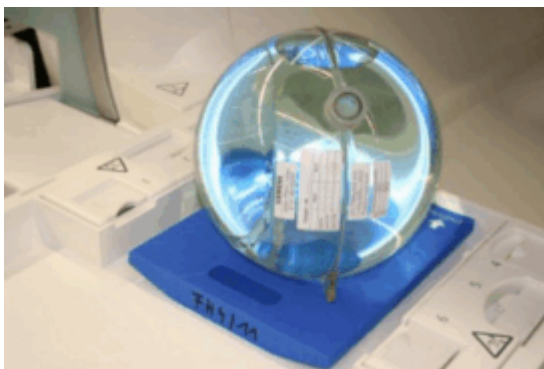
When the display cover doesn't close properly anymore after the adjustment, move plate with laser module in -Z direction, towards the magnet bore (only for Avanto).

Fig. 43: Laser beam shows on funnel



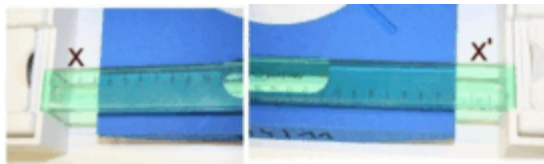
When laser beam is visible on the front funnel cover (→ Fig. 43 Page 48), move laser assembly with laser modules in +Z direction (away from the magnet bore).

Fig. 44: Ball phantom in centre



- Move the patient table to its home position and place the large ball phantom and holder (blue or red) centered on the patient table (→ Fig. 44 Page 48).

Fig. 45: Ball phantom in centre



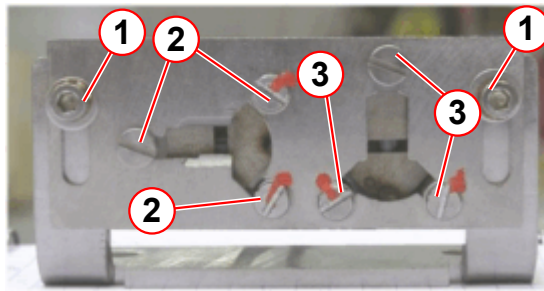
- Use a ruler and centre the arrangement by moving it in X direction until the distances X and X' are equal (→ Fig. 45 Page 48).

Fig. 46: Laser not calibrated yet



- Switch on the laser and drive the table in until the laser beams and the markers on the ball phantom meet.
- Gain access to the laser assembly by removing the light localizer cover/lid (easy to do for all Magnetoms).

Fig. 47: Laser assembly



- (1) Allen screw
- (2) Screws
- (3) Screws

- Align the two laser beams and the two markers on the ball phantom by: Backing up the two allen screws (→ 1/Fig. 47 Page 49) by a couple of turns and rotate and/or shift the entire laser module to align beams and markers on the ball phantom.

Each laser may also be adjusted individually by turning screws (→ 2/Fig. 47 Page 49) and/or (→ 3/Fig. 47 Page 49).

Fig. 48: Misaligned in X

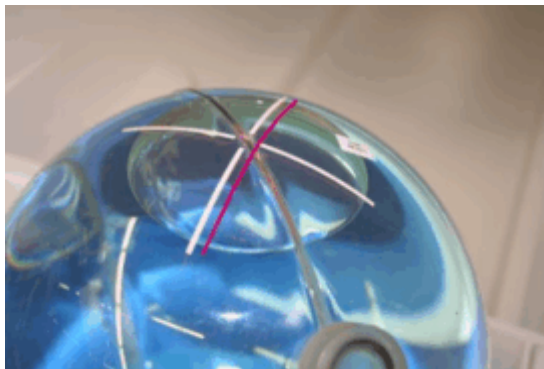
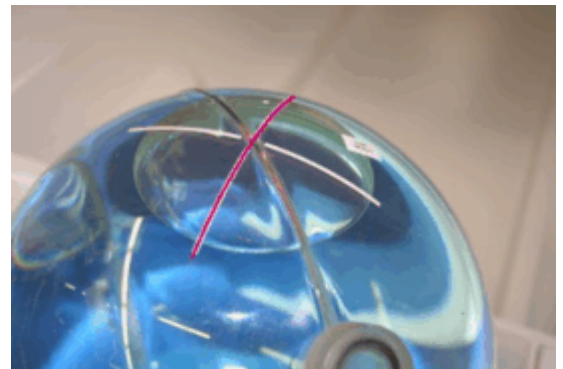
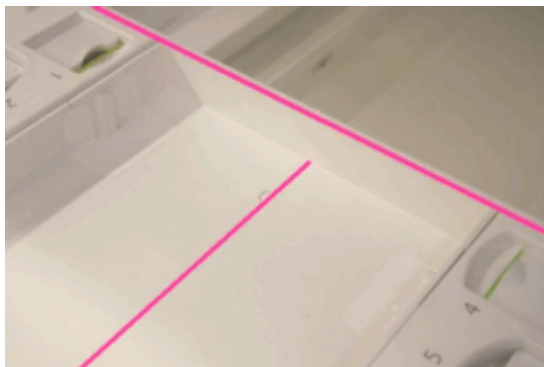


Fig. 49: X alignment with screw



- Adjust the laser element in +/- X direction by turning screw (→ 2/Fig. 47 Page 49)

Fig. 50: Aligned in X



- Move the patient table in.

The laser beam should be in parallel with the short side of the patient table tray (→ Fig. 50 Page 49). If not back up screws (→ 1/Fig. 47 Page 49) and shift/rotate entire laser assembly.

Fix the laser module with the two allen screws (→ 1/Fig. 47 Page 49) when alignment is achieved as in (→ Fig. 50 Page 49).

Fig. 51: Y-alignment



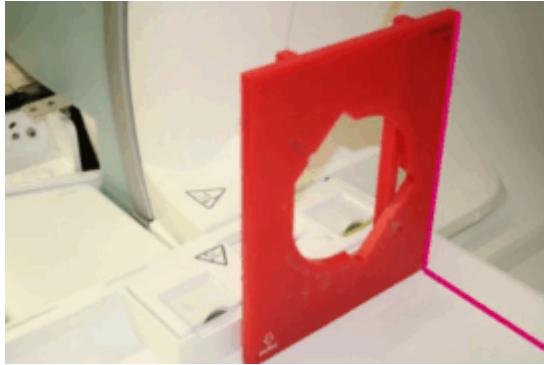
- Put one of the holders (red or blue) on the patient table as in the (→ Fig. 51 Page 50).

Fig. 52: Adjust in Y with screw C



If the laser beam is tilted in Y-direction. Turn screw (→ 3/Fig. 47 Page 49) to compensate.

Fig. 53: aligned in Y



Properly aligned in Y

3.1.2.2 Final:

- Place the large ball phantom and holder on the patient table as before.
- Turn on the laser lights.
- Move the phantom under the laser beams until they meet with the markers on the ball phantom.

Fig. 54: laser calibrated



The light localizer is calibrated when alignment is achieved as shown in (→ Fig. 54 Page 51).

4.1 Trolley

4.1.1 Replace the Engagement mechanic

The patient trolley and the patient table top are engaged during patient transportation. In case that the engagement mechanism fails it can be exchanged.

Move the trolley away from the magnet to a clear work area

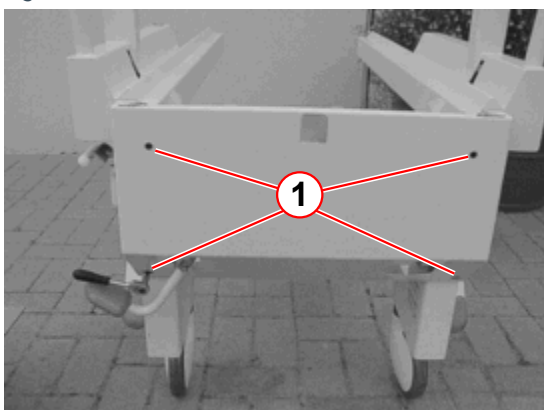
4.1.2 Special tools/Documents required

Fig. 55: Tools required



- hot air blower
- surgeon's knife

Fig. 56:



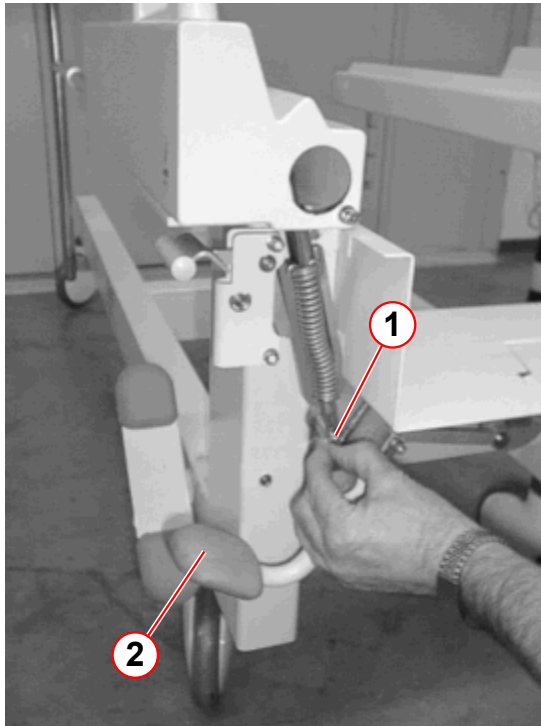
- Poke out the plastic caps with a thin tip tool (tweezers).
- Using a 5 mm allen key remove the steel cover by loosening the two upper allen screws and removing the two lower ones.

(1) 4 Allen bolts



The bolts are secured with lock tide and may be stuck in the thread hole while attempting to unscrew them. In that case use a hot air blower and direct the hot air stream at the thread, warming up the lock tide.

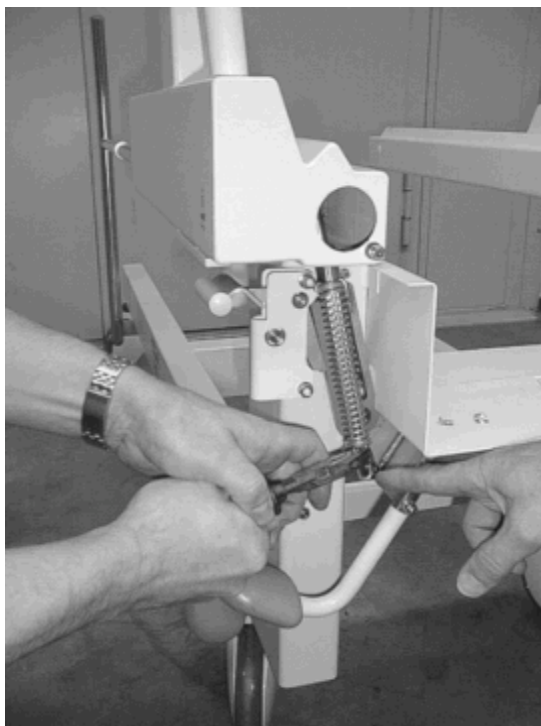
Fig. 57:



- (1) Retaining ring
- (2) Foot pedal

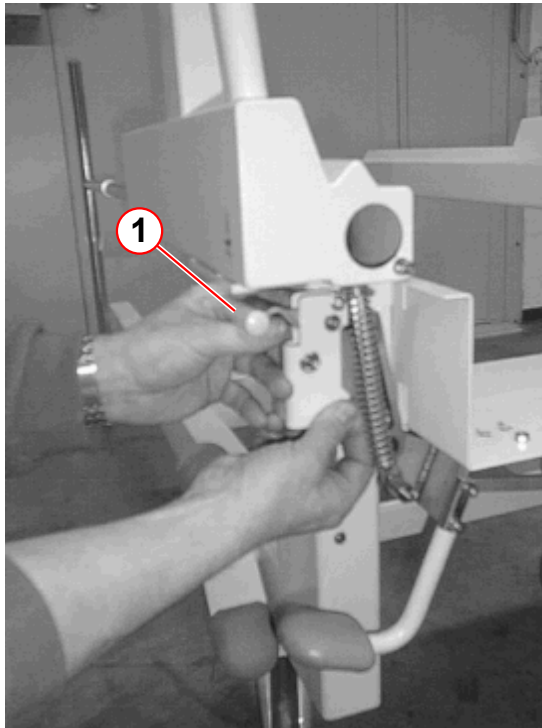
- Pull off the retaining ring for the tension spring.
 - » Remark: docking pedal can be on either side of the trolley.

Fig. 58:



- Use flat nose pliers and pull off the tension spring - get a good grip on the spring and pull it forward without applying much force.
- Let the tension spring hang loose.

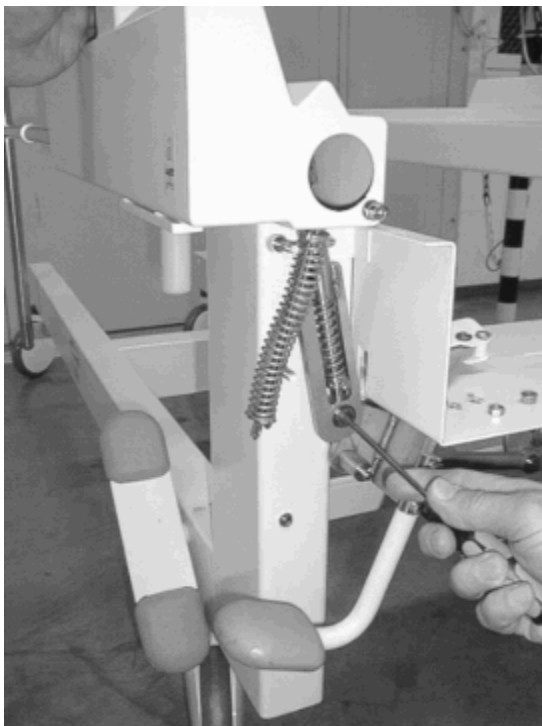
Fig. 59:



(1) Handle

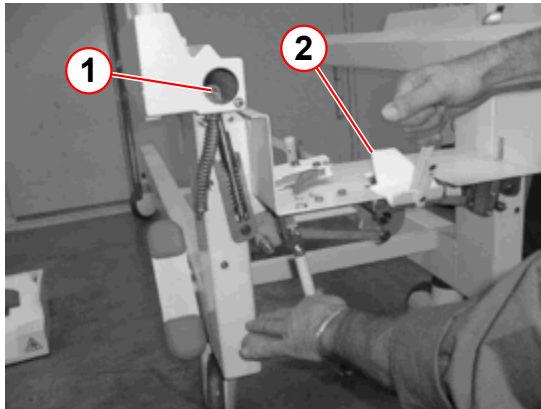
- The release handle is screwed to the table frame with two 10 mm hex-nuts.
- Remove them, take the handle off the pins and put it aside.

Fig. 60:



- With a flat blade screw-driver pull off the lower retaining ring for the damper.

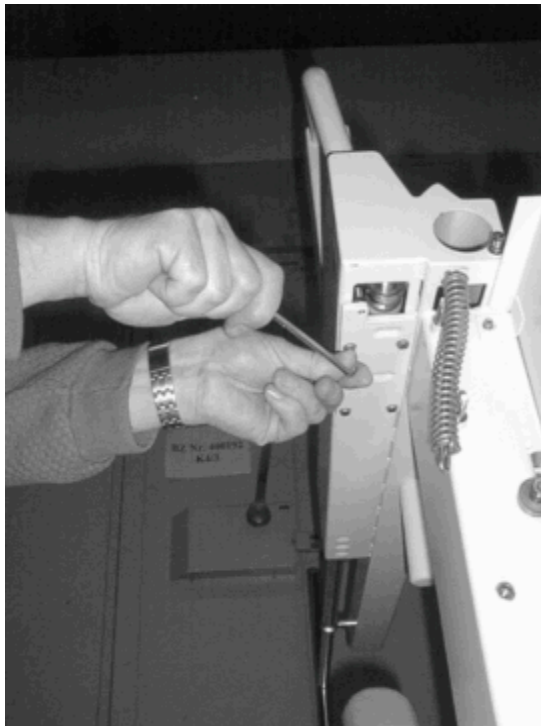
Fig. 61: Tabletop in the magnet



- (1) Upper retaining ring
- (2) Safety lever

- Pull the safety lever backwards.
- Push the foot pedal down to move the damper up so that the upper retaining ring is visible in the round window.
- With a flat blade screw-driver pull the upper retaining ring off.

Fig. 62:



- Unscrew the locking mechanism completely (5 mm allen key).

Fig. 63:

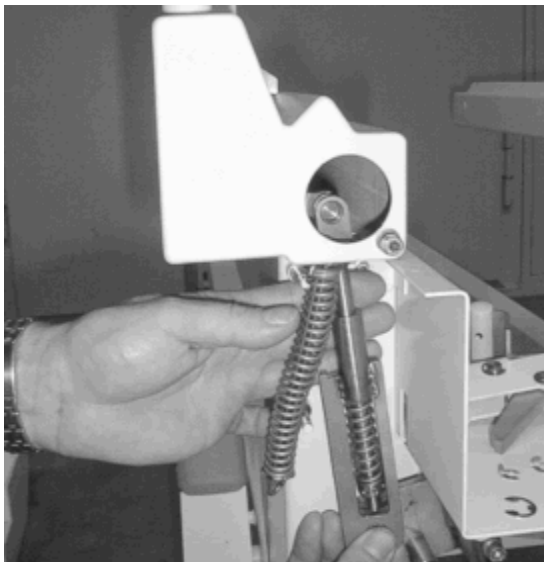


- Pull the locking mechanism down, loosen the socket set screw (3 mm allen key) on the cam lever and separate them - the locking mechanism is no longer needed.

Fig. 64:



Fig. 65:



- Grasp the damper as shown in (→ 1/ Fig. 65 Page 56) and pull it off the cam lever pin in a rocking motion.
- If necessary reach through the window with a flat nose plier and pull the damper off while holding the cam lever with the other hand.
- Put the damper aside.

4.1.3 Assembling in reversed order

Fig. 66:



- With your fingers squeeze the mechanism together, try to hold it tight or use a tie wrap and tie up the mechanism as indicated in (→ Fig. 67 Page 57).
- Feed the new locking assembly through the hole in the trolley (→ Fig. 68 Page 57), (→ Fig. 69 Page 58) and (→ Fig. 70 Page 58)).
- Cut the tie wrap with a surgeon's knife and remove it before fixing the assembly to the table.

Fig. 67:

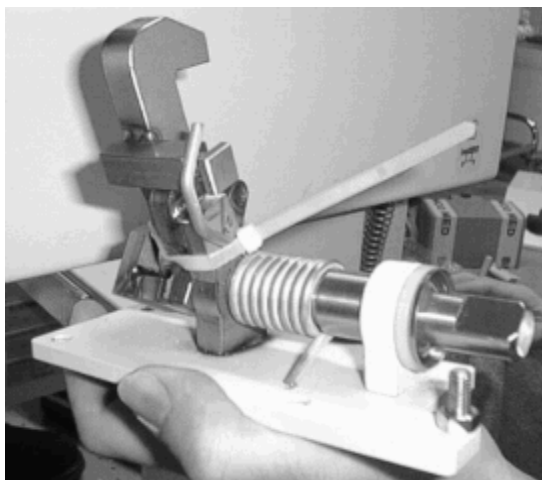


Fig. 68:

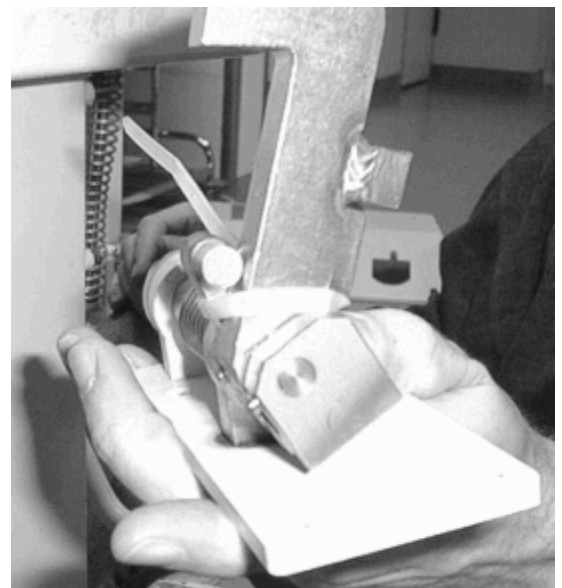


Fig. 69:



Fig. 70:

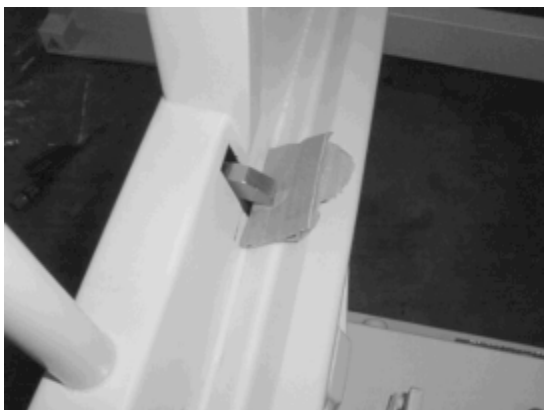


Fig. 71:



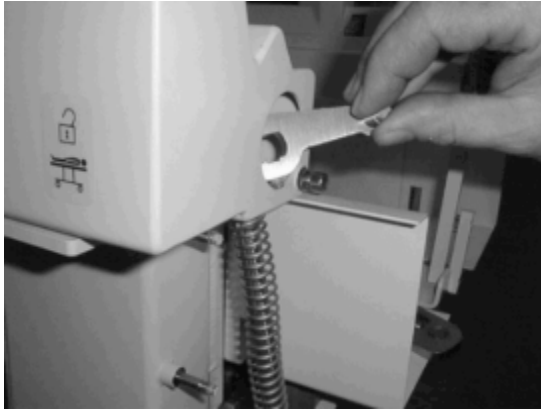
- Screw the assembly to the trolley but don't tighten yet.

Fig. 72:



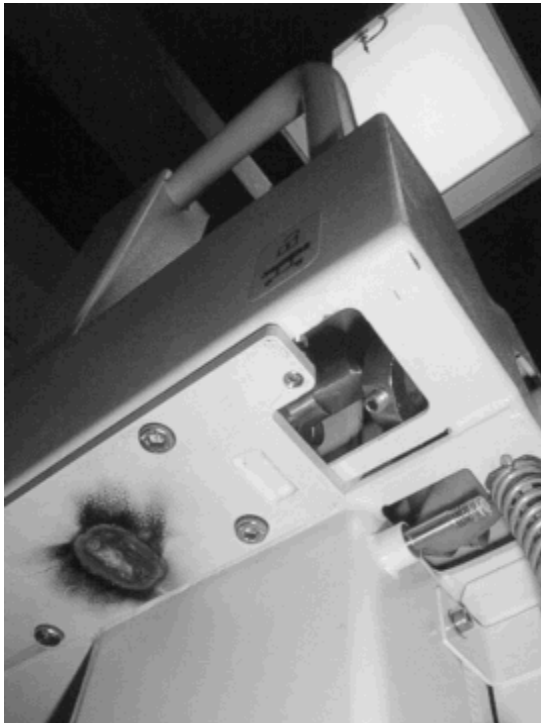
- Use a piece of card-board and squeeze it between the hook and the table.
- Centre the hook in the opening by shifting the assembly longitudinally.
- Tighten the allen bolts on the assembly.

Fig. 73:



- Feed the cam-lever through the round window with the cam in first.

Fig. 74:



- Try to slide the cam-lever onto the shaft.
- You might have to turn the shaft a little bit to align it with the cam lever.
- To do so use a pair of needle-nose pliers and reach through the hole in the table.

Fig. 75:



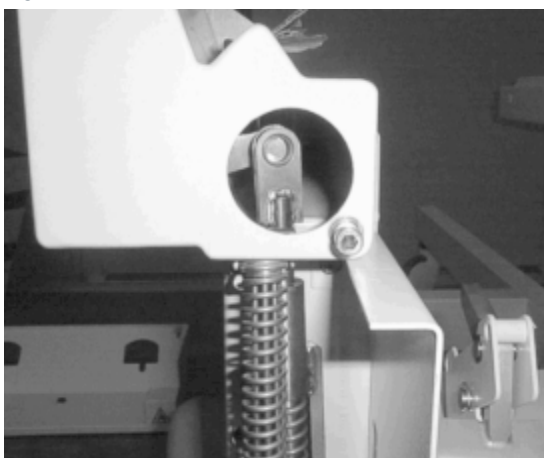
- When this is accomplished tighten the socket set screw.

Fig. 76: Damper



- » When installing the damper make sure this side goes on first.

Fig. 77:



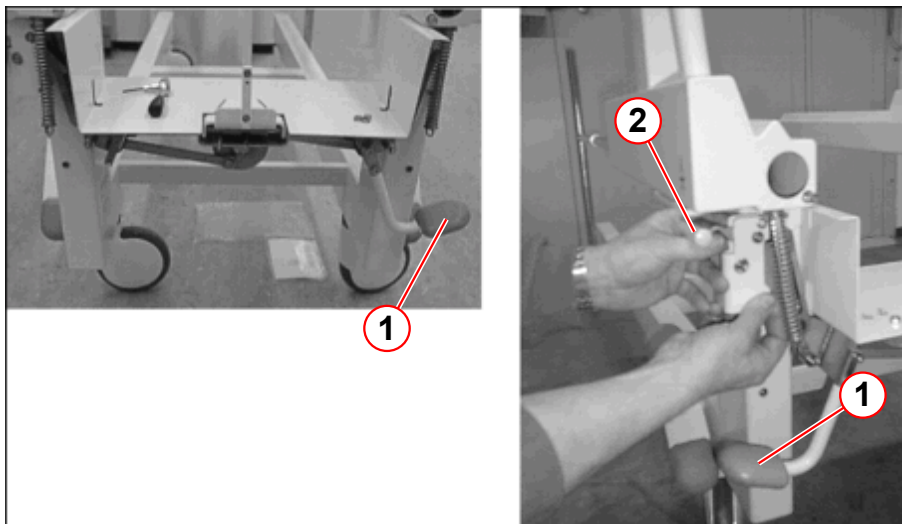
- Feed the damper over the two pins without applying much force.
- It should go on with ease.

Fig. 78:



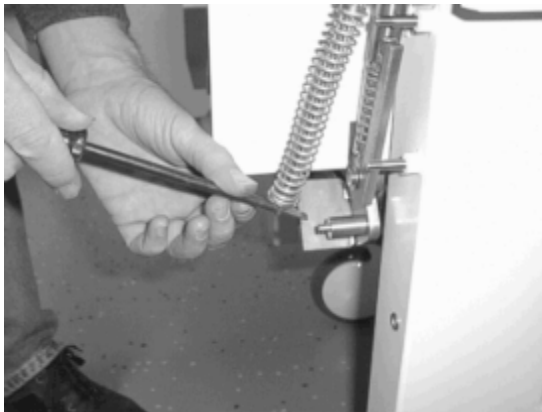
- Secure damper with new retaining rings.
- Large one on top, small one on bottom.

Fig. 79: Tabletop in the magnet



- (1) Docking pedal
(2) Handle

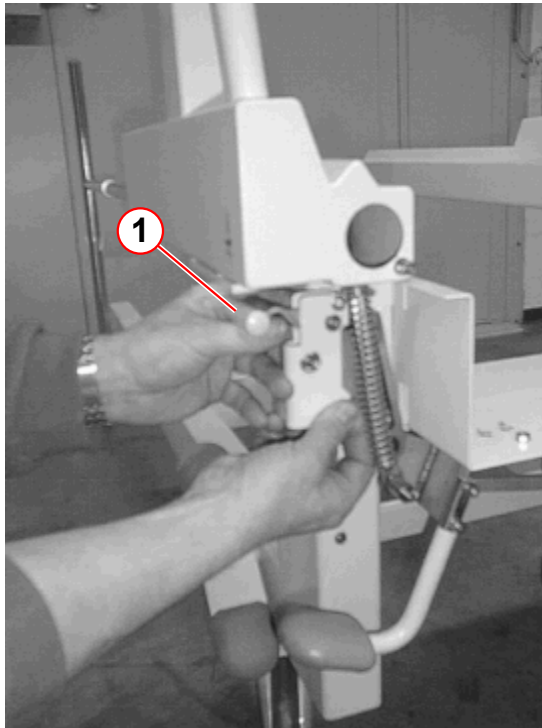
» The docking pedal should be in the unledged position.

Fig. 80:*Fig. 81:*

- Feed a screw driver through the loop on the spring.
- With force pull the spring down and over the pin.
- Pull out the screw-driver and ...
- ...push the spring loop on with your thumb.
- Secure with retaining ring.

Fig. 82:

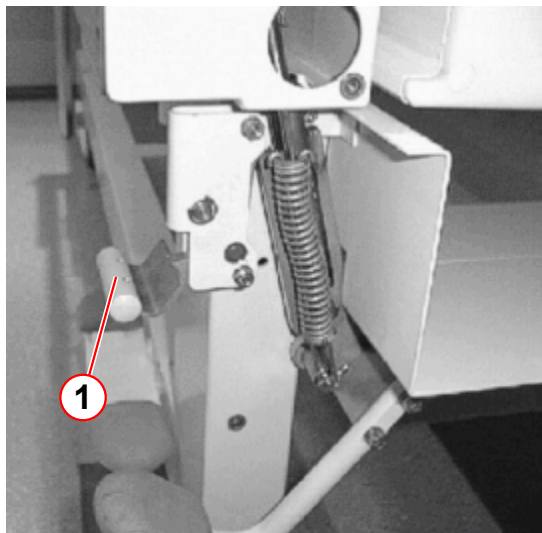
Fig. 83:



(1) Handle

- Attach release handle according to (→ Fig. 84 Page 63) with two hex nuts and washers.

Fig. 84: Release handle fitted incorrectly



(1) Release handle

Finished with one side

- » Continue with the other table side.
 - Put the rear cover back on
- For final testing perform docking and undocking cycle

- » The common print number (MR-000.841.01) from Replacement of Parts is divided into separate Component print numbers.

Version 02

- Hint: "Protective earth connection test" was added.
- Section: "Replace vertical motor assembly" was added.

Version 03

- Size of Allen key and material no. was corrected in (→ Required Tools / Page 14).
- Section: "Removing / mounting the guide rails" was corrected (Docking plate).

Version 04

- FRU: Minor editorial changes.

Version 05

- Section guide rails: Ramp down the magnet, was added.

Version 06

- Section replacing the spindle: Service bolts and hand crank added, minor editorial changes.

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